

Adaptive Plant-Health Monitoring with IoT-Informed Multi-UAV Coverage Path Planning

Research Framework (MATLAB)
Multi-UAV CPP / IPP synthesis with benchmarking

Abstract—This memorandum formalizes an adaptive forest monitoring scenario wherein a heterogeneous fleet of rotary-wing unmanned aerial vehicles (UAVs) exploits a digital plant-health heatmap fused from ground IoT cues. Unlike geometric lawn-mowing coverage, mission objectives emphasize revisitation of high-stress cells ($H \geq 3$) while respecting terrain-induced energetics and inter-agent safety margins. The novelty lies in coupling informative weights derived from health, epistemic uncertainty, and obstacle proximity with a hybrid planner that mixes weighted workload partitioning (DARP-inspired decomposition) with stitched receding-horizon A* segments over elevation-aware edge costs. Companion MATLAB modules instantiate terrain synthesis, planner comparison against greedy and boustrophedon baselines, and IEEE-oriented KPI analytics including area coverage rate (ACR), high-priority coverage (HPC), energy-to-information ratio, and cross-track RMSE versus a canonical sweep reference.

I. SCENARIO AND GRID MODEL

Let $\mathcal{G} = \{1, \dots, N\} \times \{1, \dots, M\}$ denote a uniform planar decomposition with cell spacing Δ . Each cell $i \in \mathcal{G}$ carries discrete stress $H_i \in \{0, \dots, 4\}$, sensing uncertainty $\sigma_i \in (0, 1]$, obstacle occupancy $b_i \in \{0, 1\}$, and obstacle proximity score $O_i \in [0, 1]$ (high near obstacles). Traversable cells satisfy $b_i = 0$. Terrain elevation $z_i \in \mathbb{R}$ affects climb penalties and altitude-dependent power draw.

II. PRIORITY FUNCTIONAL

The planner consumes a scalar IoT fusion weight

$$W_i = \alpha H_i + \beta \sigma_i - \gamma O_i, \quad (1)$$

with nonnegative coefficients α, β, γ . High-priority targets emphasize stressed vegetation ($H_i \geq \tau$), uncertain telemetry (σ_i), while penalizing proximity to collision hazards.

III. MULTI-OBJECTIVE PROBLEM STATEMENT

Let $\mathcal{U} = \{1, \dots, U\}$ index UAVs executing discrete-time trajectories $\mathbf{p}_u(t) \in \mathcal{G}$ with corresponding continuous velocities $v_u(t)$ and flight altitudes $h_u(t) = h_{\text{AGL}} + z(\mathbf{p}_u(t))$. Denote $\mathcal{V}(t)$ the set of cells visited by any agent up to time t , A_i the planar cell area, and $I_i(t)$ an information increment when cell i is observed with diminishing novelty encoded via σ_i . We pursue Pareto refinement of three competing objectives:

- 1) maximize cumulative coverage utility $J_{\text{cov}} = \sum_{i \in \mathcal{V}(T)} \Psi(W_i) A_i$ with monotone Ψ (identity in implementation);

- 2) minimize energy $J_{\text{ene}} = \sum_{u \in \mathcal{U}} \int_0^T P(v_u, h_u) dt$ for a nonnegative powerModel P ;
- 3) maximize terminal information $J_{\text{inf}} = \sum_{t \leq T} \sum_u \eta(\mathbf{p}_u(t))$, $\eta = \sigma_i$ on first visitation of i .

IV. AGGREGATED COST FUNCTIONAL

For nonnegative weights $\lambda_1, \lambda_2, \lambda_3$, compose a weighted-sum surrogate amenable to decentralized replanning:

$$J = -\lambda_1 \underbrace{\frac{1}{T} \sum_t \dot{A}_{\text{cov}}(t)}_{\text{ACR}} + \lambda_2 \underbrace{\frac{J_{\text{ene}}}{\max(\varepsilon, J_{\text{inf}})}}_{\text{energy-to-information}} - \lambda_3 J_{\text{inf}} \\ \approx -\lambda_1 \text{ACR} + \lambda_2 \text{EGI} - \lambda_3 J_{\text{inf}}, \quad (2)$$

where $\dot{A}_{\text{cov}}$ approximates instantaneous newly covered area rate (m^2/s) and EGI denotes energy gain inefficiency (Joules per unit accumulated σ -weighted novelty).

V. POWER AND ENERGY MODEL

Adopting a parsimonious multicopter drag-and-induced-power surrogate,

$$P(v, h) = P_0 + k_v v^3 + k_h h, \quad P \geq P_{\min}, \quad (3)$$

with calibrated constants $\{P_0, k_v, k_h, P_{\min}\}$ mapping commanded horizontal speed v and absolute altitude h to electrical shaft/load power.

VI. CONSTRAINTS

A. Kinematics

Grid-adjacent motion respects $\|\mathbf{p}_u(t+1) - \mathbf{p}_u(t)\|_{\infty} \leq 1$, translating to speed bound $v_u \leq v_{\max}$ under spacing Δ and sampling period Δt .

B. Collision Avoidance

Pairwise safety demands $\|\mathbf{x}_u(t) - \mathbf{x}_{u'}(t)\|_2 \geq d_{\text{safe}}$ with $\mathbf{x}$ denoting metric embeddings ($\text{col} \cdot \Delta$, $\text{row} \cdot \Delta$).

C. Obstacle Feasibility

$$\mathbf{p}_u(t) \in \mathcal{F} = \{i \in \mathcal{G} : b_i = 0\} \text{ for all } t, u.$$

VII. PLANNER SYNOPSIS

A. Baseline Lawnmower

Deterministic serpentine enumeration of $\mathcal{F}$ partitioned across UAVs.

B. Baseline Greedy

Myopic eight-connected expansion maximizing local W plus coverage novelty bonuses.

C. Priority Hybrid

Weighted Lloyd partitioning on coordinates of $\{i : H_i \geq \tau\}$ using replication proportional to W_i , followed by weighted nearest-neighbor goal sequencing and elevation-aware A* stitching; optional bounded-window shortcuts emulate receding-horizon refinement.

VIII. EVALUATION PROTOCOL

A. Test Matrix

Monte Carlo seeds sweep **terrains** $\in \{\text{flat, hilly, ridge}\}$ crossed with **tree densities** $\in \{\text{low, medium, high}\}$ producing obstacle fields of increasing occupancy.

B. Metrics

$$\text{ACR} = \frac{1}{T} \sum_t \frac{A_{\text{new}}(t)}{\Delta t} \quad [\text{m}^2/\text{s}], \quad (4)$$

$$\text{HPC} = 100 \cdot \frac{|\{i : H_i \geq \tau\} \cap \mathcal{V}(T)|}{|\{i : H_i \geq \tau\} \cap \mathcal{F}|} \quad [\%], \quad (5)$$

$$\text{EGI} = \frac{J_{\text{ene}}}{\sum_i \sigma_i \mathbb{I}_{\text{first visit}}(i)} \quad [\text{J}], \quad (6)$$

$$\text{RMSE}_{\text{ct}} = \sqrt{\frac{1}{T} \sum_t d^2(\mathbf{x}_1(t), \mathcal{L}_{\text{lm}})} \quad [\text{m}], \quad (7)$$

$$\text{EPB} = \frac{J_{\text{ene}}}{\sum_i b_i \mathbb{I}_{\text{first visit}}(i)} \quad [\text{J/bit}], \quad (8)$$

where $\mathcal{L}_{\text{lm}}$ is the polyline of the lawnmower reference for UAV 1, and b_i denotes an expected information contribution in bits (Gaussian surrogate).

IX. CONSTRAINED OPTIMUM FOR PRECISION AGRICULTURE

For regulatory-style crop stressing audits one reports mission statistics at coverage quorum τ (e.g. 80% HPC). The constrained planning surrogate is

$$f^* = \min_{\mathbf{x}} E_{\text{total}}(\mathbf{x}) \quad \text{s.t.} \quad \text{HPC}(\mathbf{x}) \geq \tau, \quad (9)$$

where $\mathbf{x}$ stacks planner tunings (fusion weights, capacitated Voronoi rounds, sortie energy caps). Empirically compare f^* against a *fair* lawnmower baseline that integrates energy only until the same HPC threshold is met (`MissionSim.stopHpcPct`), while optionally annotating full-grid lawnmower energy for operational context. Multi-objective analysis overlays the unconstrained energy-information Pareto hull among feasible designs.

X. EXTENSIONS (ACTIVE SENSING, BUDGETED SORTIES, HETEROGENEOUS DECOMPOSITION)

A. GP Surrogate and Expected Information

Interpreting σ_i as a marginal posterior standard deviation on latent health, an approximate entropy reduction for a noisy Gaussian observation with noise variance σ_n^2 is

$$\text{EIG}_i \approx \frac{1}{2} \log_2(1 + \sigma_i^2/\sigma_n^2) \quad [\text{bits}]. \quad (10)$$

High- H actionable cells prioritize EIG_i for active sensing blends beyond linear W .

B. Mission Objective Under Sortie Budgets

Let $\mathcal{S}_k$ index sorties with recharge inserts of duration Δt_{rc} . Minimize total clock time $\sum_k T_k + \Delta t_{\text{rc}}$ subject to per-sortie fleet energy caps $\bar{E}_k$ until $\text{HPC} \geq 90\%$.

C. Three-Axis Power Split

$$P = \underbrace{P_0(1 + \kappa_p v^2)}_{\text{blade/induced}} + \underbrace{k_h |v|^3 + k_a v^2}_{\text{horizontal}} + \underbrace{k_{\uparrow} (\dot{z}_+)^2 + k_{\downarrow} (\dot{z}_-)^2}_{\text{terrain}} + k_z h, \quad (11)$$

with terrain rate $\dot{z}$ coupling uphill asymmetry.

D. Heterogeneous Spatial Decomposition

Capacitated Voronoi iterations partition targets across UAV seeds using workload feedback λ_u , reducing contention versus unstructured hotspots-only clustering.

XI. IMPLEMENTATION MAP

`MapGenerator.m` synthesizes (z, H, σ, b, O, W) plus `GP/EIG` attachments (`GPInformationField.m`); `PriorityPlanner.m` implements heuristic/EIG blends with Voronoi or Lloyd decomposition and optional Chaikin smoothing (`PathSmoother.m`); `BatteryAwareOrchestrator.m` executes multi-pass missions; `MissionSim.m` integrates the split power law, bit budgets, optional intrasortie IoT drift; `Analytics.m` adds EPB aggregates; `runParetoSensitivity.m` / `runBatteryAwarePHM.m` illustrate sweeps; `runParetoEnergyInformation.m` renders the $\text{HPC} \geq \tau$ constrained energy-information figure; `runAdaptivePHM_Benchmark.m` retains the 3×3 matrix.

REFERENCES

- [1] Framework references should cite domain CPP, DARP, IPP, and field robotics trials aligned with project bibliography policies.